Name

Mathematics Advanced Year 11

Reciprocal Functions

Book 1

Version: 261021

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

Reciprocal functions

- Graph functions of the form $f(x) = \frac{k}{x}$, where k is a constant and $k \neq 0$, and identify their hyperbolic shape and their asymptotes
- Describe the behaviour of $f(x) = \frac{k}{x}$ as $x \to \infty$ and $x \to -\infty$

Hyperbolas

ⓘ Reciprocal Relationships

Reciprocal relationships are in the form $y = \frac{k}{x}$.

Their graph will not be a straight line; so the relationship is **non-linear.**

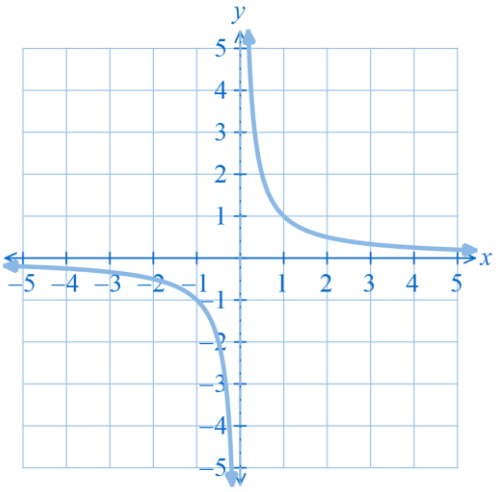
We can graph reciprocal relationships using these steps:

1. Construct a table of values using the equation.
2. Graph the table by using the columns as coordinates.
3. Join the points to form a smooth curve.

🔍 **Investigation** Graph reciprocal relationships using a table of values

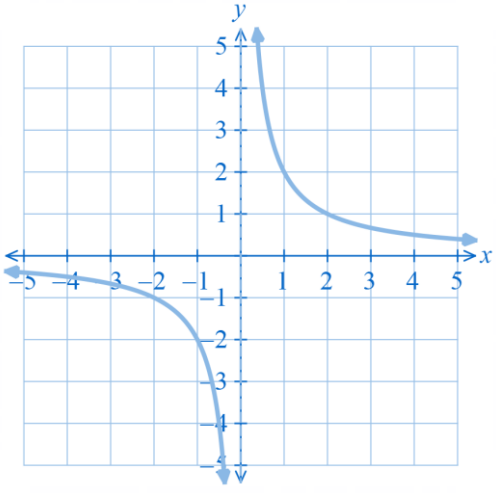
Draw the graph of $y = \frac{1}{x}$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							



Draw the graph of $y = \frac{2}{x}$

x	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2
y							

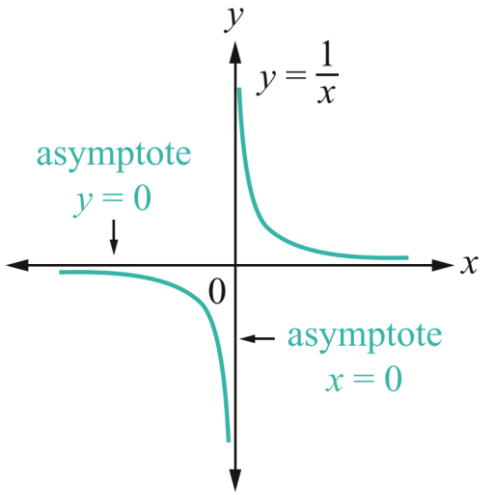


ⓘ Properties of Hyperbolas

A **reciprocal** relationship $y = \frac{k}{x}$ has a graph in the shape of a **hyperbola**.

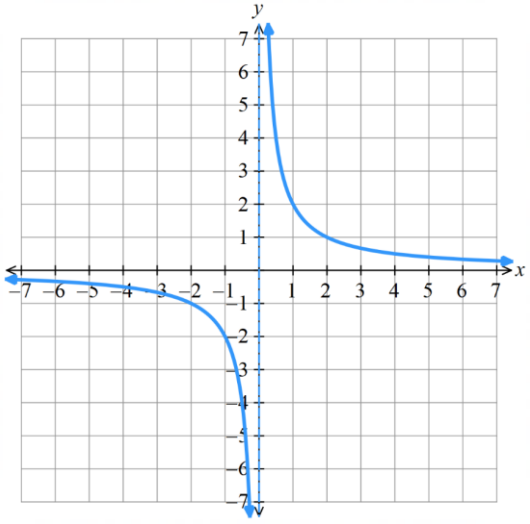
Hyperbolas have:

- Two **asymptotes**: lines where the graph approaches but never reaches.
- No x or y-intercepts.
- Two symmetrical branches.



Identify Features of hyperbolas

a $y = \frac{2}{x}$



Identify the asymptotes.

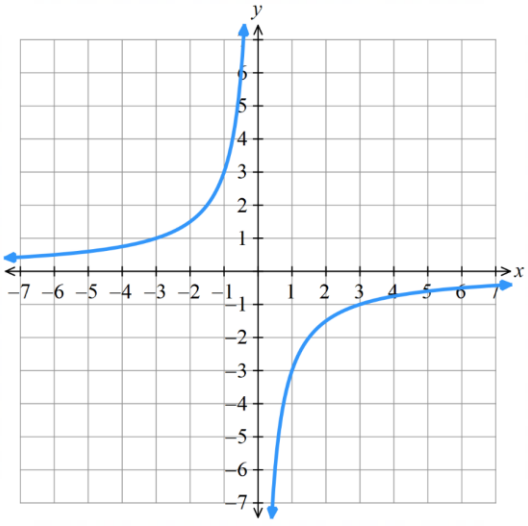
Behaviour as $x \to \infty$:

Behaviour as $x \to -\infty$:

Behaviour as $x \to 0^+$:

Behaviour as $x \to 0^-$:

b $y = -\frac{2}{x}$



Identify the asymptotes.

Behaviour as $x \to \infty$:

Behaviour as $x \to -\infty$:

Behaviour as $x \to 0^+$:

Behaviour as $x \to 0^-$:

Investigation Equation and the graph of a hyperbola

https://www.desmos.com/calculator

a Sketch these graphs

$y = \frac{1}{x}$

$y = \frac{2}{x}$

$y = \frac{3}{x}$

$y = \frac{4}{x}$

Increasing the value of k s…… or *dilates* the hyperbola away from the axes.

b Sketch these graphs.

$y = \frac{-1}{x}$

$y = \frac{-2}{x}$

$y = \frac{-3}{x}$

$y = \frac{-4}{x}$

A negative value of k means the hyperbola is in the …… and …… quadrants.

c Sketch these graphs

$y = \frac{1}{x} + 1$

$y = \frac{1}{x} + 2$

$y = \frac{1}{x} + 3$

$y = \frac{1}{x} - 3$

$y = \frac{1}{x} - 4$

The constant term shifts the hyperbola v……

d Sketch these graphs

$y = \frac{1}{x+1}$

$y = \frac{1}{x+2}$

$y = \frac{1}{x+3}$

$y = \frac{1}{x-3}$

$y = \frac{1}{x-4}$

The constant term with the x shifts the hyperbola h……

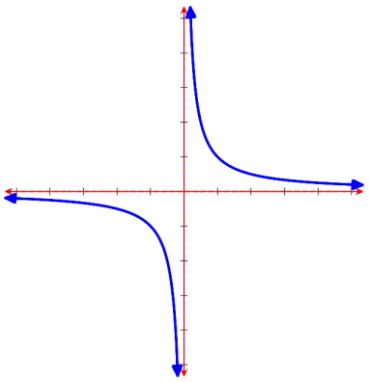
ⓘ Hyperbola Transformations

Consider the equation of a hyperbola $y = \frac{k}{x-b} + c$, where k is an integer.

The parameters k, b, c are numbers that affect the graph in predictable ways.

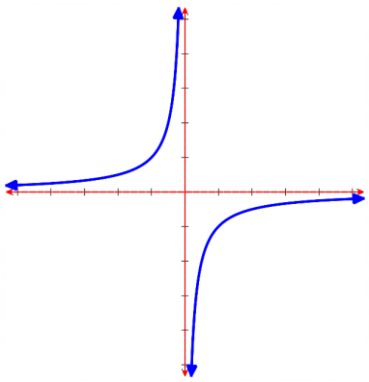
k is positive, the hyperbola is top …… and bottom …… relative to the asymptotes.

$y = \frac{1}{x}$



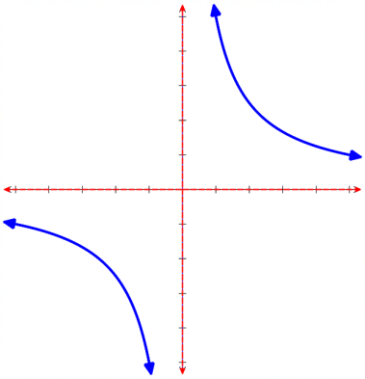
k is negative, the hyperbola is …… and …… relative to the asymptotes.

$y = -\frac{1}{x}$

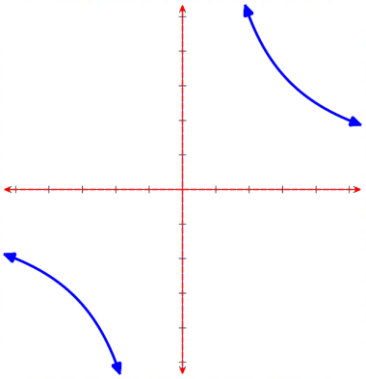


As $|k|$ increases, the hyperbola gets *dilated*; it stretches away from the axes.

$y = \frac{5}{x}$

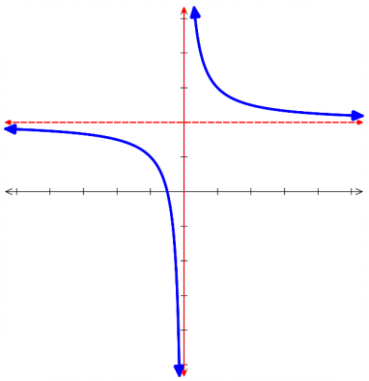


$y = \frac{10}{x}$

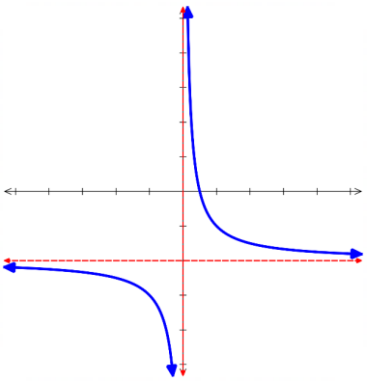


c shifts the curve …… The value of c is the …… asymptote.

$y = \frac{1}{x} + 2$

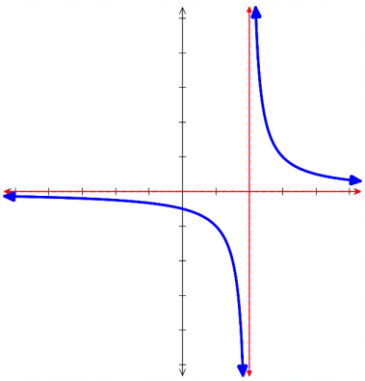


$y = \frac{1}{x} - 2$

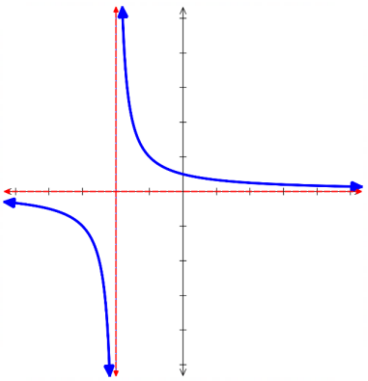


b shifts the curve …… The value of b is the …… asymptote.

$y = \frac{1}{x-2}$



$y = \frac{1}{x+2}$



Sketching Hyperbolas

Dilates, or stretches, hyperbola

$$y = \frac{k}{x-b} + c$$

Horizontal translation: $x = b$ is a vertical asymptote

Vertical translation: $y = c$ is a horizontal asymptote

To sketch hyperbolas:

1. Find y-intercept, if any, by letting $x = 0$.
2. Find x-intercept, if any, by letting $y = 0$.
3. Find vertical asymptote at $x = b$ and horizontal asymptote at $y = c$,

Example Sketch hyperbolas, showing intercepts and asymptotes

$$y = \frac{1}{x+1} + 2$$

1. y-int: Let $x = 0$,

$$y = \frac{1}{1} + 2 = 3$$

2. x-int: Let $y = 0$

$$0 = \frac{1}{x+1} + 2$$

$$-2 = \frac{1}{x+1}$$

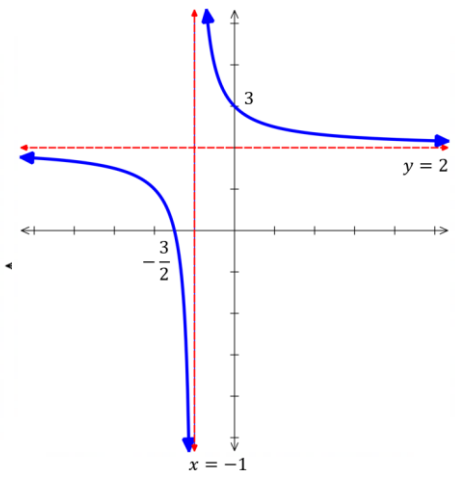
$$-2x - 2 = 1$$

$$-2x = 3$$

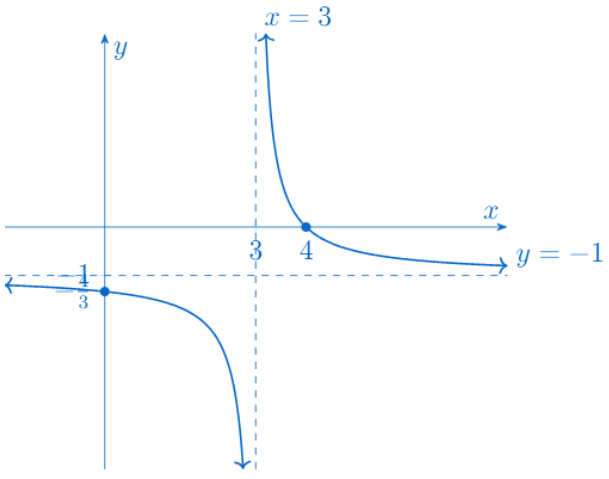
$$x = -\frac{3}{2}$$

3. Asymptotes: $x = -1$

$y = 2$



$$y = \frac{1}{x-3} - 1$$



ⓘ Alternate Form of a Hyperbola

Another form of a hyperbola is

$$y = \frac{px + q}{rx + s}$$

We can convert this to the standard form by splitting the numerator.

Example Sketch hyperbolas in alternate form.

$y = \frac{2x+3}{x-1}$

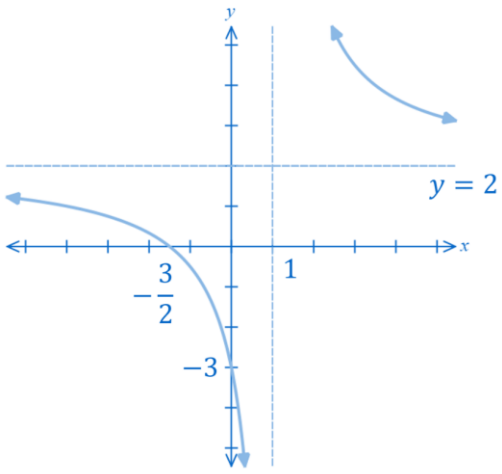
0. Convert to standard form

$$y = \frac{2x + 3}{x - 1}$$
$$= \frac{2x - 2 + 2 + 3}{x - 1}$$
$$= \frac{2x - 2}{x - 1} + \frac{5}{x - 1}$$
$$= 2 + \frac{5}{x - 1}$$

1. y-int: Let $x = 0$,
$y = 2 + \frac{5}{-1} = -3$

2. x-int: Let $y = 0$
$$0 = 2 + \frac{5}{x - 1}$$
$$-2 = \frac{5}{x - 1}$$
$$-2x + 2 = 5$$
$$-2x = 3$$
$$x = -\frac{3}{2}$$

3. Asymptotes: $x = 1$
$y = 2$



$y = \frac{3x-1}{x+2}$

